

Hazelnut-enriched diet improves cardiovascular risk biomarkers beyond a lipid-lowering effect in hypercholesterolemic subjects.

BACKGROUND: Tree nuts, particularly almonds, walnuts, and pistachios, have been shown to possess cardioprotective effects. However, there is little information on the effects of hazelnut consumption on cardiovascular risk markers. **METHODS:** The antiatherogenic effect of hazelnut before and after consumption in hypercholesterolemic subjects was investigated. Twenty-one hypercholesterolemic volunteers (18 men and 3 women) were recruited in a double control sandwich model intervention study with a single group and three isoenergetic diet periods. These were control diet I (4 weeks), hazelnut-enriched diet (4 weeks; hazelnut contributing 18%-20% of the total daily energy intake), and control diet period II (4 weeks). The cardiovascular risk biomarkers such as endothelial function, using flow-mediated dilation (FMD) technique, low-density lipoprotein (LDL) oxidation products and inflammatory markers such as high-sensitivity C-reactive protein (hs-CRP), soluble intercellular adhesion molecule-1, and soluble vascular cell adhesion molecule-1 (sVCAM-1) as well as lipids and lipoprotein levels were monitored. **RESULTS:** Consumption of a hazelnut-enriched diet significantly improved FMD (56.6%), total cholesterol (-7.8%), triacylglycerol (-7.3%), LDL-cholesterol (-6.17%), and high-density lipoprotein cholesterol (6.07%) compared with the control diet I. Oxidized-LDL, hs-CRP, and sVCAM-1 levels were significantly lower in the group ingesting a hazelnut-enriched diet compared with the control diets I and II. Modest correlations between sVCAM-1 and FMD and between sVCAM-1 and hs-CRP were observed ($r = -0.49$, $P < .025$; $r = 0.66$, $P < .001$, respectively). **CONCLUSION:** Hazelnut-enriched diets may exert antiatherogenic effect by improving endothelial function, preventing LDL oxidation, and inflammatory markers, in addition to their lipid and lipoprotein-lowering effects. These beneficial effects appeared to be reversible after 4 weeks on a hazelnut-free diet. Therefore, hazelnut may be incorporated into daily diet without change in total caloric intake for sustained health benefit. Copyright © 2013 National Lipid Association. All rights reserved.